

p_T resolution fit vs $|\eta^{\text{truth}}|$: r_0 free vs $r_0 = 0$

$$\sigma_{p_T}/p_T = \sqrt{r_0^2/p_T^2 + r_1^2 + (r_2 \times p_T)^2} \quad -- \quad \sigma_{68} = (q_{84} - q_{16})/2 \quad -- \quad Z + Z' \text{ combined}$$

$ \eta $ bin	N points	r_0 free [GeV]	χ^2/ndf (r_0 free)	χ^2/ndf ($r_0 = 0$)	r_2 [GeV^{-1}]
$0.0 \leq \eta < 0.1$	16	0.000 ± 36.336	0.58	0.51	$0.224 \pm 0.012 \ (\times 10^{-3})$
$0.1 \leq \eta < 1.05$	16	0.000 ± 8.987	1.55	1.35	$0.126 \pm 0.008 \ (\times 10^{-3})$
$1.05 \leq \eta < 1.3$	15	0.000 ± 36.854	0.49	0.43	$0.124 \pm 0.009 \ (\times 10^{-3})$
$1.3 \leq \eta < 1.7$	14	0.000 ± 42.829	1.75	1.53	$0.159 \pm 0.011 \ (\times 10^{-3})$
$1.7 \leq \eta < 2.5$	13	0.000 ± 31.452	0.65	0.57	$0.089 \pm 0.008 \ (\times 10^{-3})$
$2.5 \leq \eta < 2.8$	11	0.221 ± 0.337	0.09	0.09	$0.104 \pm 0.010 \ (\times 10^{-3})$

In red: $\chi^2/\text{ndf} > 3$, unreliable fit

In blue: r_2 , the nominal ATLAS resolution at high p_T